

BHARATH SIMHA REDDY BHUREDDY

AI Engineer | Production LLM Systems & Microservices

✉ bharathsimhareddybhureddy@gmail.com | 📞 +91-7670978187 | 🐙 GitHub | 🔗 LinkedIn

Summary

AI Engineer focused on turning models into products. I build production-ready LLM systems, RAG pipelines, and scalable model serving architectures using Python, FastAPI, and Docker to solve real-world business problems.

Skills

Programming	: Python, SQL
ML & DL	: PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib
GenAI & LLMs	: LangChain, LangGraph, RAG Pipelines, Fine-tuning (LoRA), Hugging Face, Gradio, Sentence Transformers, FAISS, Pinecone
Backend	: FastAPI, Pydantic, PostgreSQL, MongoDB, Supabase, REST APIs
DevOps	: Docker, GitHub Actions, CI/CD, Linux, Git, Pytest, Locust
Observability	: LangSmith, Langfuse, DeepEval
Cloud	: AWS ECS, GCP Vertex AI

Work Experience

Bodega Delivery – AI/ML Engineering Intern 🐙 **Jul 2025 – Jan 2026 | Guntur, AP**

- Pitched and built a real-time FastAPI app using Gemini Vision API. Delivery drivers upload photos right before handoff to verify food freshness and expiry dates, reducing poor-quality deliveries and customer complaints by 20%.
- Eliminated the need to manually read through 60+ business documents and restaurant menus across 10 B2B clients. Engineered a RAG pipeline utilizing BM25 retrieval to instantly query and extract specific client details, cutting information discovery time by 50%.

Projects

Career AI – AI-Powered Career Coach | 🐙 Repo **2026**

- Engineered a custom async FastAPI backend using LLMs and strict Pydantic models to extract structured data from PDFs, improving extraction accuracy by 30% over traditional parsers.
- Deployed containerized architecture to AWS ECS, reducing API response time by 40% through `asyncio.gather` concurrent processing and model caching.
- Validated system robustness using Locust, sustaining 80+ requests per second and stabilizing 95th-percentile latency under a 200 concurrent user load.

LLM Fine-Tune & Eval Engine | 🐙 Repo **2026**

- Fine-tuned a LLaMA-2 model on college study PDFs to create a domain-expert academic assistant, reducing hallucinations on curriculum-specific queries.
- Engineered an automated evaluation suite using DeepEval (LLM-as-a-Judge) to measure faithfulness, building a Gradio dashboard that demonstrated a 25% reduction in hallucinations versus the base model.

Deepfake Detection System | 🐙 Repo **2025**

- Trained a MobileNet transfer-learning model on 190,000 images to classify synthetic versus real media, achieving an 85% baseline accuracy.
- Deployed the detection system as a full-stack application, utilizing FastAPI for efficient, low-latency model serving and Gradio for an interactive user interface.

Education

Annamacharya Institute of Technology and Sciences **2022 – 2026**

B.Tech in Artificial Intelligence and Machine Learning *CGPA: 8.43/10*

Achievements

- **2nd Place, AITS Codethon:** Secured runner-up out of 18 competing teams in an intra-college algorithmic problem-solving and coding competition.
- **Technical Writing & Education:** Actively authoring the "Zero to Production AI Engineer" guide, creating step-by-step documentation to teach scalable ML deployment and backend architecture.